

VARUN SREENIVASAN

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EXPERIENCE

DataChat, Inc.

Senior Software Engineer

Madison, WI

Feb 2022 - Present

- Achieved consistently high performance ratings from team leads across quarterly assessments.
- Proposed and implemented adoption of the [Artillery](#) load testing tool to evaluate performance and scalability of DataChat's SaaS application. Testing efforts helped uncover system bottlenecks that impeded application performance under load.
- Led push towards CI/CD to streamline release management process involving upgrades of Kubernetes hosted environments. This effort helped increase upgrade cadence and reduce manual effort.
- Accelerated LLM adoption across various application use cases by setting up an evaluation framework for benchmarking. In addition, led the research to assess feasibility of hosting DataChat's custom LLMs across multiple cloud providers and platforms.
- Spearheaded design and implementation of backend solutions to expand product capabilities within ML & GenAI space and satisfy customer demands. Followed best system design practices to achieve scalability, compute efficiency, and reliability.

National Science Foundation – IRIS HEP

[Fellow](#) – Graph Methods for Particle Tracking (High Luminosity Large Hadron Collider)

Berkeley, CA

May 2021 - August 2021

- Developed embedding pipeline to find distance metric between 3D hit measurements pairs, optimizing particle differentiation.
- Accelerated the nearest neighbors search by replacing Facebook's Faiss with Fixed Radius Nearest Neighbors (FRNN) on CUDA.

Citrine Informatics

[NextGen-Fellow](#) – Computational Materials Science

Redwood City, CA

May 2018 - August 2018

- Multi-university research project: Competitively selected, successfully completed bootcamp & workshop at Stanford University, presented findings at NextGen Research Symposium in Golden, CO, and co-authored a paper (see publications section).
- Employed RetinaNet object detection model to identify defects in metals; Developed evaluation pipeline to determine recall and precision metrics; Performed hyperparameter optimization to obtain model with 85% precision and 68% recall on the test set.

EDUCATION

M.S. Computer Science, University of Wisconsin-Madison (Dec 2021)

G.P.A: 4.00/4.00

B.S. Computer Science (Minor: Mathematics), University of Wisconsin-Madison (May 2020)

G.P.A: 3.90/4.00 "Distinction in the Major" and Dean's List

LANGUAGES & TECHNOLOGY SKILLSETS

Python, Git, AWS, GCP, Azure, Terraform, Docker, Kubernetes, Helm, CircleCI, VS Code, Java, Jupyter Notebook, Unix, MacOS

LICENSES & CERTIFICATIONS

Amazon Web Services: Solutions Architect Associate, Cloud Practitioner, Machine Learning Specialty

OTHER SOFTWARE PROJECTS

Autonomous Vehicles

Built proof of concept for an open source autonomous vehicle testbed by training PyTorch based SSD Mobilenet object detection model to do live detection of traffic signs. The proposal was accepted and presented at ACM MobiArch'22 (see publications section).

Business Success/Viability Forecast

Developed multiple ML models using Yelp dataset to predict whether businesses can withstand impacts of COVID-19. Tasks involved performing feature engineering, creating training and validation pipelines, and determining the vital factors affecting business survival.

RELEVANT COURSEWORK

Machine Learning, Data Science, Artificial Intelligence, Algorithms, Computer Networks, Mobile & Wireless Networks, Computer Vision, Cryptography, Combinatorics, Linear Algebra & Differential Equations, Operating Systems.

PUBLICATIONS & PRESENTATIONS

- [μCity: a miniaturized autonomous vehicle testbed](#): Procs. 17th ACM Workshop **MobiArch'22**
- [Multi defect detection & analysis of electron microscopy images with deep learning](#): Science Direct (Computational Material Sci.)
- [Deep learning for automated defect detection in TEM images - RetinaNet](#): Citrine Informatics fellowship final presentation
- [Graph methods for particle tracking](#): IRIS HEP fellowship final presentation